

Subject-Independent EEG Mental-Workload Classification on STEW

Baselines · artifact analysis · augmentation · proposed artifact-robust training (ACCT)

Table 1. Key numbers at a glance.

Result	Value
Best baseline (Spectral + LR), balanced accuracy	0.829 [0.796, 0.861]
Best deep baseline (ATCNet)	0.803 $\pm$ 0.010
Best model without F7/F8/T7/T8 (ATCNet)	0.783
Largest augmentation gain (EEG Conformer, 12 subjects, CWGAN-GP)	+8.2 points
ACCT vs. standard training, EEGNet (validation subjects)	0.807 vs. 0.768 (+3.8, p_Holm = 0.035)
Prediction flips under injected artifacts $\times 3$	24.5% $\rightarrow$ 7.0%
Calibration error (ECE)	0.064 $\rightarrow$ 0.030

1. Data and protocol

Item	Setting
Dataset	STEW: 48 subjects, Emotiv EPOC, 14 channels, 128 Hz; rest vs. SIMKAP multitasking, 150 s each
Channels	AF3 F7 F3 FC5 T7 P7 O1 O2 P8 T8 FC6 F4 F8 AF4
Filtering	0.5–45 Hz, 4th-order Butterworth, zero-phase
Windows	4 s (512 samples), 50% overlap
Artifact rejection	Peak-to-peak > median + 60 $\times$ MAD, per subject, class-agnostic
Normalisation	z-score per window and channel (no per-recording normalisation: would leak the label)
Split	10-fold cross-validation by subject; each subject tested once; 5 seeds
Validation	15% of training subjects (subject-disjoint), early stopping only
Deep training	AdamW, lr 1e-3, batch 64, cosine schedule + warm-up, $\leq$ 100 epochs, same recipe for all
Metrics	Balanced accuracy (primary), accuracy, macro-F1, κ , MCC, ROC-AUC, PR-AUC, sensitivity, specificity, Brier, log-loss, ECE; window and recording level
Statistics	Subject-cluster bootstrap 95% CI; Friedman + Nemenyi; Wilcoxon signed-rank + Holm
Hardware	NVIDIA RTX A4000 16 GB, NGC PyTorch 26.07 container

2. Baseline benchmark (25 models)

Table 2. Subject-independent results, mean $\pm$ SD over 5 seeds. † uses unlabelled test-subject data (transductive).

Model	Family	BAcc	95% CI	AUC	κ	ECE	Rec. Acc
Spectral + LR	Feature-based	0.829	[0.796, 0.861]	0.904	0.658	0.022	0.938
Spectral + SVM (RBF)	Feature-based	0.829	[0.791, 0.865]	0.903	0.657	0.018	0.906
Spectral + sLDA	Feature-based	0.827	[0.795, 0.858]	0.905	0.655	0.061	0.958
Riemann re-centred TS + LR †	Riemannian	0.825	[0.778, 0.865]	0.895	0.650	0.029	0.979
Spectral + RF	Feature-based	0.821 $\pm$ 0.002	[0.786, 0.854]	0.897	0.641	0.097	0.927
Spectral + GBDT	Feature-based	0.805	[0.769, 0.840]	0.893	0.609	0.037	0.917
ATCNet	EEG transformer	0.803 $\pm$ 0.010	[0.764, 0.841]	0.882	0.605	0.062	0.898
EEGNet-8,2	EEG CNN	0.801 $\pm$ 0.014	[0.763, 0.839]	0.872	0.602	0.044	0.900
EEG-TCNet	EEG CNN	0.797 $\pm$ 0.004	[0.757, 0.836]	0.877	0.594	0.054	0.879
Riemann TS + LR	Riemannian	0.795	[0.757, 0.831]	0.875	0.590	0.040	0.885
TSception	EEG CNN	0.794 $\pm$ 0.006	[0.756, 0.830]	0.877	0.587	0.030	0.885
ShallowConvNet	EEG CNN	0.785 $\pm$ 0.012	[0.750, 0.819]	0.860	0.570	0.057	0.871
1D-CNN	Generic CNN	0.784 $\pm$ 0.004	[0.747, 0.819]	0.855	0.567	0.049	0.875
EEG Conformer	EEG transformer	0.778 $\pm$ 0.005	[0.739, 0.815]	0.858	0.553	0.135	0.873
CNN-LSTM	Generic CNN	0.776 $\pm$ 0.012	[0.742, 0.811]	0.831	0.552	0.071	0.873
GRU	Recurrent	0.766 $\pm$ 0.005	[0.730, 0.801]	0.830	0.531	0.057	0.854
BiLSTM	Recurrent	0.765 $\pm$ 0.005	[0.731, 0.798]	0.827	0.529	0.045	0.871
LSTM	Recurrent	0.763 $\pm$ 0.014	[0.732, 0.792]	0.824	0.525	0.035	0.902
DGCNN	Graph	0.761 $\pm$ 0.010	[0.725, 0.796]	0.832	0.522	0.047	0.885
Swin (STFT)	Vision transformer	0.740 $\pm$ 0.013	[0.706, 0.774]	0.817	0.480	0.082	0.844
ViT (STFT)	Vision transformer	0.734 $\pm$ 0.008	[0.699, 0.768]	0.805	0.468	0.066	0.827
Electrode GCN	Graph	0.724 $\pm$ 0.009	[0.685, 0.762]	0.799	0.446	0.034	0.819
Riemann MDM	Riemannian	0.713	[0.665, 0.762]	0.783	0.426	0.191	0.729
Vanilla RNN	Recurrent	0.703 $\pm$ 0.009	[0.677, 0.730]	0.753	0.407	0.034	0.867
DeepConvNet	EEG CNN	0.633 $\pm$ 0.087	[0.591, 0.678]	0.769	0.268	0.216	0.656

Classical models are deterministic given the folds ($SD = 0$), except RF. Rec. Acc = accuracy after averaging each 150 s recording.

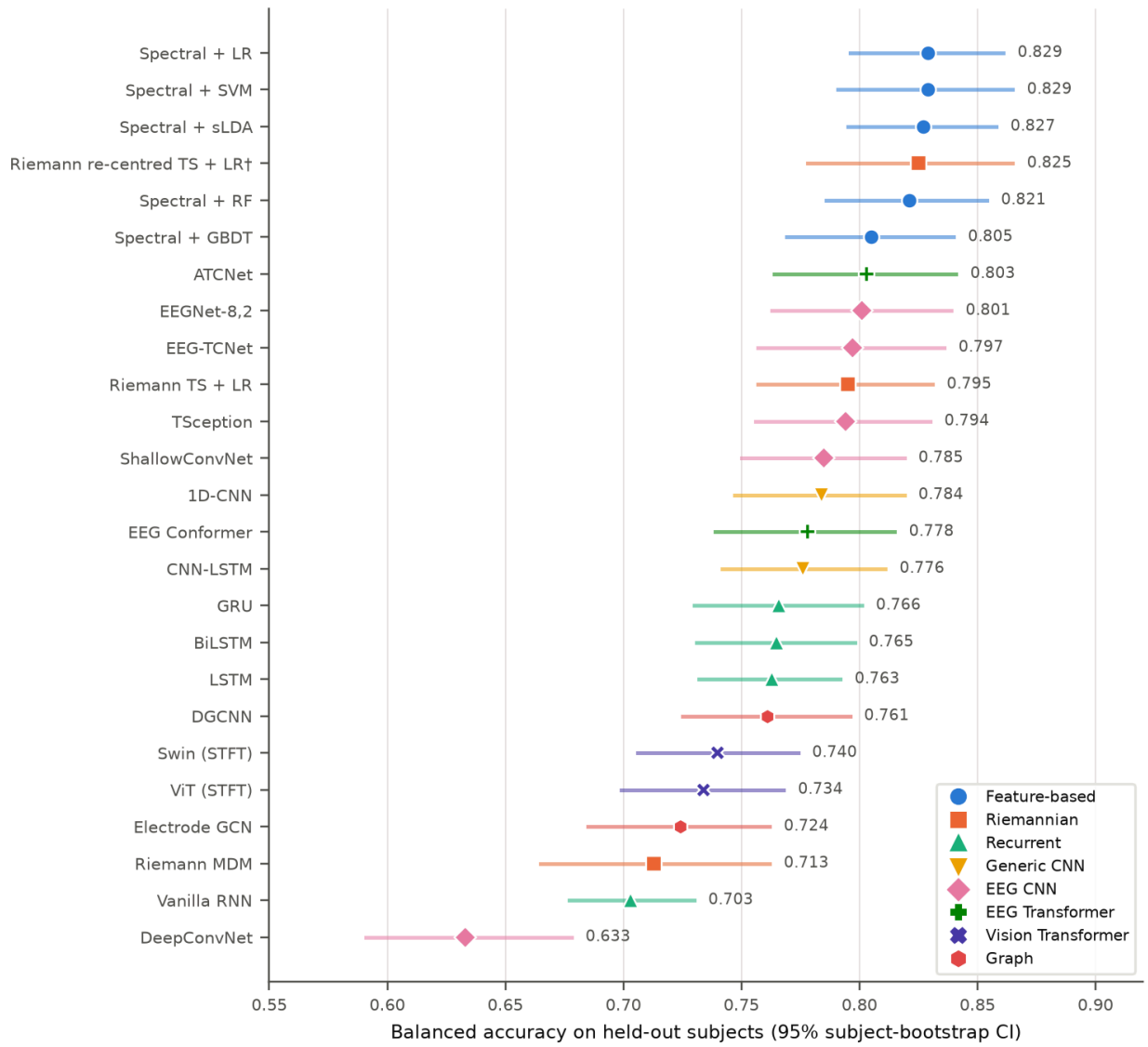


Figure 1. Balanced accuracy with 95% subject-bootstrap CIs, by model family.

- **Friedman:** $\chi^2 = 283.0$, $p = 4.3 \times 10^{-46}$; Nemenyi CD = 5.50 ranks.
- **Wilcoxon vs. best (Holm):** significantly worse only for MDM, vanilla RNN, LSTM, DeepConvNet, ViT, Swin, electrode GCN, DGCNN; top EEG CNNs and ATCNet not significantly different.
- **Overfitting:** several deep models peak at validation epoch 1–3 (ShallowConvNet, 1D-CNN, ViT, TSception).
- **Calibration:** feature models ECE ≈ 0.02 ; EEG Conformer over-confident (0.135); DeepConvNet unstable across seeds.

Table 3. Model cost (one 4 s window; GPU latency at batch size 1).

Model	Parameters	MFLOPs	GPU latency (ms)	Train time / fold (s)
EEGNet-8,2	1,842	7.7	0.23	10.3
EEG-TCNet	3,814	2.2	0.46	11.5
TSception	7,838	24.2	0.29	18.1
ShallowConvNet	27,682	29.7	0.10	6.8
ATCNet	112,650	11.8	2.01	22.8
BiLSTM	140,546	141.6	1.29	23.2
EEG Conformer	745,506	44.4	0.89	19.2

3. Task vs. rest band power

Table 4. Largest task – rest differences in log band power (paired t, n = 48, BH-FDR over 70 tests).

Channel	Band	t	d_z	q (FDR)
F8	theta	+8.61	+1.24	2.2×10^{-9}
FC6	theta	+6.68	+0.96	8.8×10^{-7}
F8	delta	+5.87	+0.85	9.9×10^{-6}
F8	gamma	+5.78	+0.83	1.0×10^{-5}
T8	theta	+5.59	+0.81	1.3×10^{-5}
AF4	theta	+3.93	+0.57	1.5×10^{-3}
P8	alpha	-3.02	-0.44	0.016
O2	alpha	-2.88	-0.42	0.021

- **Expected workload pattern:** frontal theta ↑, posterior alpha ↓.
- **Artifact sign:** F7, F8, T8 rise in every band (delta–gamma), typical of eye/muscle activity.

4. Artifact ablation

Table 5. Balanced accuracy when artifact-prone information is removed (3 seeds, same folds).

Model	All channels	No gamma	No F7/F8/T7/T8	Both removed	Δ channels
Spectral + SVM	0.829	0.825	0.778	0.785	-5.1
Spectral + LR	0.829	0.824	0.777	0.776	-5.2
Spectral + sLDA	0.827	0.819	0.770	0.769	-5.7
Riemann TS + LR	0.795	0.799	0.776	0.777	-1.9
ATCNet	0.803	0.797	0.783	0.777	-2.0
EEGNet-8,2	0.801	0.795	0.780	0.770	-2.1
EEG-TCNet	0.797	0.789	0.775	0.769	-2.2
TSception	0.794	0.805	0.780	0.770	-1.4
ShallowConvNet	0.785	0.786	0.768	0.759	-1.7

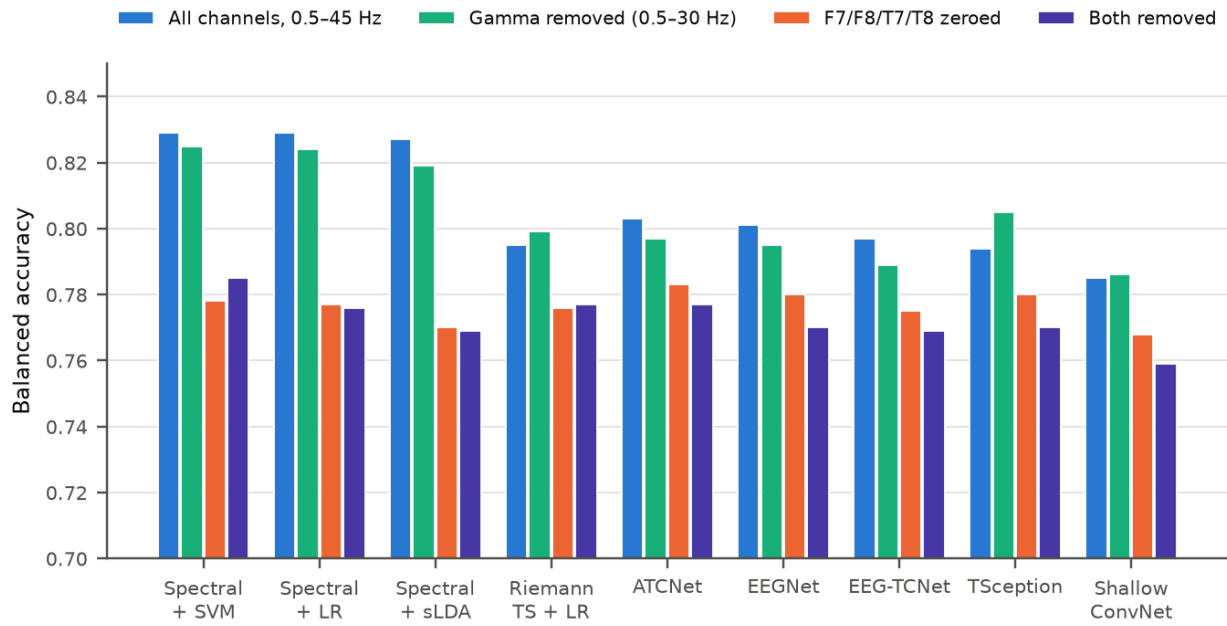


Figure 2. Effect of removing gamma and the eye-adjacent channels.

- **Gamma:** ≤ 1 point change $\rightarrow$ muscle activity is not the main factor.
- **F7/F8/T7/T8:** spectral models lose 5–6 points, deep models 1.4–2.2; ranking flips to deep models (ATCNet 0.783).
- **Still ~ 0.78 without them:** most of the decodable signal is neural.

5. Augmentation: standard vs. GAN vs. diffusion

8 online augmentations, CWGAN-GP and conditional DDPM; generators trained inside each fold on training subjects only; 3 seeds.

Table 6. All training subjects. Δ = paired change vs. none (* Holm $p < 0.05$).

Augmentation	EEG Conformer	EEGNet-8,2	ShallowConvNet
None	0.776	0.801	0.794
Channel dropout	0.801 (+0.027*)	0.807 (+0.007)	0.810 (+0.017)
Mixup	0.798 (+0.023*)	0.795 (−0.006)	0.797 (+0.004)
Gaussian noise	0.792 (+0.017*)	0.801 (+0.001)	0.793 (−0.000)
CWGAN-GP, 50% extra	0.787 (+0.012)	0.778 (−0.021*)	0.779 (−0.015)
CWGAN-GP, 100% extra	0.782 (+0.008)	0.768 (−0.033*)	0.769 (−0.025*)
DDPM, 50% extra	0.770 (−0.004)	0.804 (+0.004)	0.793 (+0.000)
DDPM, 100% extra	0.780 (+0.005)	0.790 (−0.010)	0.799 (+0.007)

Table 7. Balanced accuracy by number of training subjects per fold (6 / 12 / 24 / all).

Augmentation	EEG Conformer	EEGNet-8,2	ShallowConvNet
None	.632 / .668 / .773 / .776	.712 / .742 / .775 / .801	.733 / .743 / .774 / .794
CWGAN-GP 100%	.661 / .752 / .764 / .782	.710 / .727 / .738 / .768	.735 / .756 / .777 / .769
DDPM 100%	.641 / .733 / .773 / .780	.703 / .727 / .763 / .790	.726 / .772 / .780 / .799
Mixup	.679 / .698 / .780 / .798	.707 / .727 / .770 / .795	.710 / .746 / .775 / .797
Channel dropout	.657 / .674 / .778 / .801	.717 / .745 / .780 / .807	.745 / .752 / .788 / .810
Frequency shift	.698 / .686 / .762 / .789	.732 / .745 / .773 / .800	.714 / .738 / .772 / .788

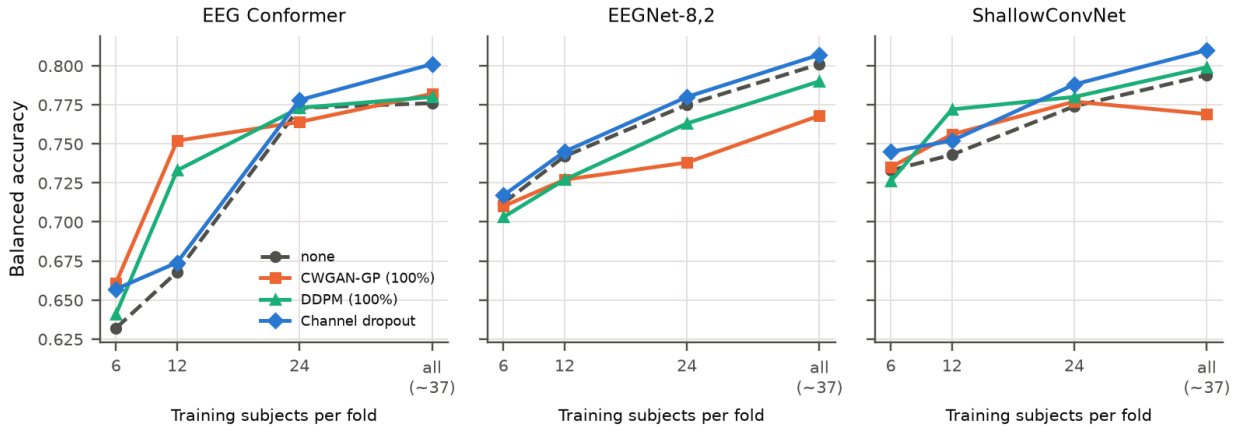


Figure 3. Accuracy vs. training subjects per fold.

Table 8. Synthetic-data quality (mean over folds, seeds, classes). TSTR = train synthetic, test real; TRTR = real $\rightarrow$ real.

Subjects	Generator	LSD (dB)	MMD	Density	Coverage	Memor.	TSTR	TRTR
6	CWGAN-GP	2.79	0.074	0.648	0.325	0.981	0.603	0.690
6	DDPM	3.92	0.321	0.446	0.138	0.989	0.647	0.690
12	CWGAN-GP	2.71	0.046	0.732	0.365	0.952	0.582	0.713
12	DDPM	3.65	0.141	1.081	0.367	0.931	0.688	0.713
24	CWGAN-GP	2.85	0.037	1.068	0.488	0.935	0.599	0.747
24	DDPM	4.67	0.211	0.985	0.403	0.915	0.709	0.747
all	CWGAN-GP	2.55	0.034	1.278	0.572	0.941	0.600	0.760
all	DDPM	5.12	0.246	0.997	0.453	0.907	0.707	0.760

- **Few subjects (12):** synthetic data helps the transformer: Conformer +8.2 (CWGAN), +6.4 (DDPM); ShallowConvNet + DDPM +3.0.
- **All subjects:** CWGAN hurts EEGNet (−3.3) and ShallowConvNet (−2.5); DDPM neutral; channel dropout most consistent.
- **Fidelity $\neq$ usefulness:** CWGAN matches spectra better (lower LSD/MMD) but carries less class information (TSTR 0.60 vs. 0.71).

6. Proposed method: ACCT

Artifact-counterfactual consistency training: the classifier is trained to give the same prediction for a window and for a copy with eye/muscle artifacts added or removed. No extra sensor (EOG) needed; inference is unchanged.

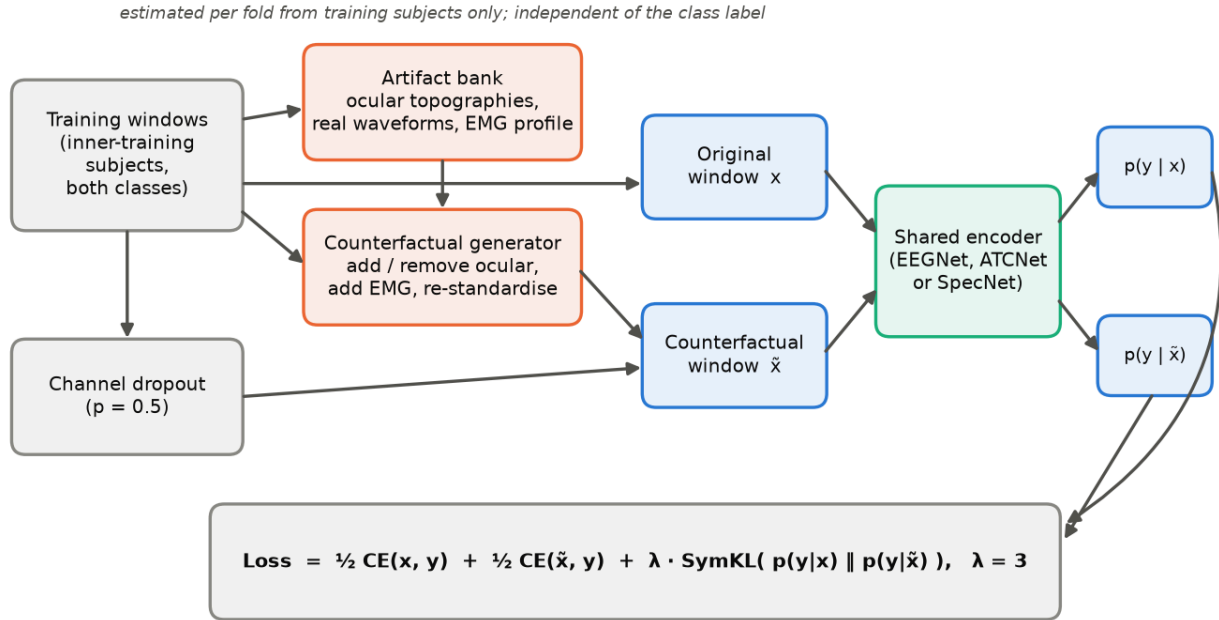


Figure 4. ACCT training scheme.

Table 9. Components.

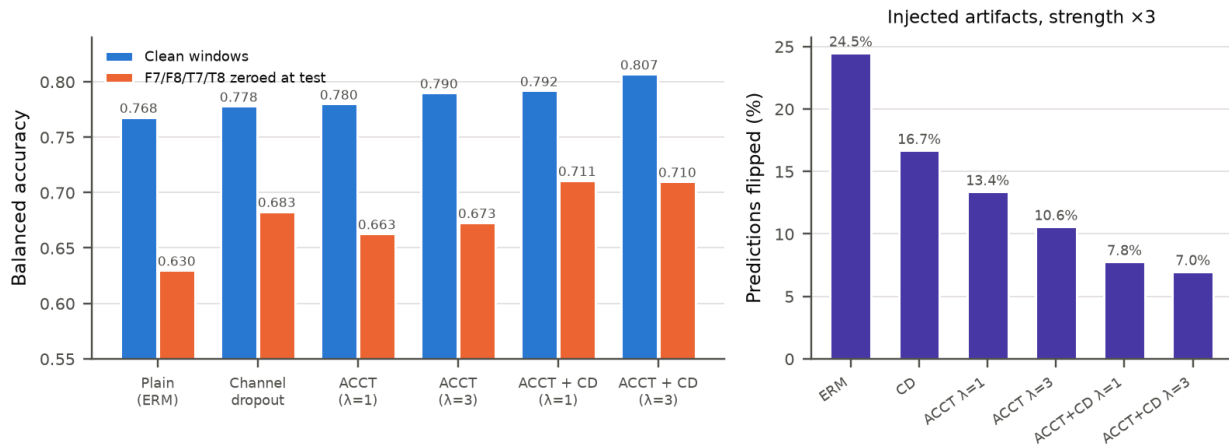
Component	What it does
Artifact bank	Per fold, training subjects only, both classes pooled. Ocular: frontal-dominant principal components of < 4 Hz activity + bank of real waveforms. EMG: channel profile of excess 20–45 Hz power.
Counterfactual generator	Per window, label-independent: add ocular ($p = 0.5$), remove ocular subspace ($p = 0.3$), add EMG bursts ($p = 0.5$); re-z-scored.
Consistency loss	$L = \frac{1}{2} \text{CE}(\mathbf{x}, \mathbf{y}) + \frac{1}{2} \text{CE}(\tilde{\mathbf{x}}, \mathbf{y}) + \lambda \cdot \frac{1}{2} [\text{KL}(p(\mathbf{x}) \parallel p(\tilde{\mathbf{x}})) + \text{KL}(p(\tilde{\mathbf{x}}) \parallel p(\mathbf{x}))]$, $\lambda = 3$
Channel dropout	Whole channels zeroed at random during training.
Encoder	Any network. EEGNet for development; ATCNet, TSception, EEG-TCNet and SpecNet (differentiable spectral model, ~7k parameters) in round 3.

Table 10. Development results, EEGNet, validation subjects only (3 seeds). CD = channel dropout.

Training	Clean BAcc	AUC	ECE	BAcc F7/F8/T7/T8 = 0	BAcc high-ocular	Natural gap
Standard (ERM)	0.768 ± 0.007	0.836	0.064	0.630	0.698	+0.107
Channel dropout	0.778 ± 0.020	0.849	0.058	0.683	0.710	+0.105
Counterfactuals only	0.768 ± 0.007	0.839	0.061	0.635	—	—
ACCT $\lambda = 1$	0.780 ± 0.012	0.854	0.056	0.663	0.716	+0.099
ACCT $\lambda = 3$	0.790 ± 0.011	0.860	0.046	0.673	0.726	+0.099
ACCT + CD $\lambda = 1$	0.792 ± 0.002	0.871	0.032	0.711	0.718	+0.120
ACCT + CD $\lambda = 3$	0.807 ± 0.008	0.880	0.030	0.710	0.730	+0.117

Table 11. Injected-artifact stress test: balanced accuracy / predictions flipped.

Training	Strength $\times 1$	Strength $\times 2$	Strength $\times 3$
Standard (ERM)	0.752 / 7.6%	0.707 / 17.5%	0.667 / 24.5%
Channel dropout	0.771 / 4.8%	0.748 / 11.0%	0.723 / 16.7%
Counterfactuals only	0.766 / 4.2%	0.752 / 9.7%	0.732 / 15.1%
ACCT $\lambda = 1$	0.776 / 3.7%	0.765 / 8.2%	0.744 / 13.4%
ACCT $\lambda = 3$	—	—	0.761 / 10.6%
ACCT + CD $\lambda = 1$	—	—	0.778 / 7.8%
ACCT + CD $\lambda = 3$	—	—	0.793 / 7.0%

**Figure 5.** Clean vs. channel-ablated accuracy (left); flips under injected artifacts (right).**Table 12.** Paired per-subject tests vs. standard training (Wilcoxon, Holm).

Training	Metric	Δ	p_Holm
ACCT + CD $\lambda = 3$	Clean BAcc	+0.038 [+0.017, +0.060]	0.035
ACCT + CD $\lambda = 3$	BAcc, F7/F8/T7/T8 zeroed	+0.079	0.00059
ACCT + CD $\lambda = 1$	BAcc, F7/F8/T7/T8 zeroed	+0.081	1.1×10^{-6}
ACCT $\lambda = 3$	BAcc, F7/F8/T7/T8 zeroed	+0.043	0.0025
ACCT $\lambda = 1$	Clean BAcc	+0.012	0.14 (n.s.)
Counterfactuals only	Clean BAcc	≈ 0	n.s.

- **Active ingredient:** the consistency term (counterfactuals alone give no clean gain).
- **Best configuration:** ACCT + channel dropout, $\lambda = 3$.
- **Limitations:** stress test uses the same artifact family as training; natural gap (most vs. least eye-active windows) unchanged at ≈ 0.10 – 0.12 ; validation subjects only, test run pending.

7. Status

Item	Status
25-model benchmark, neurophysiology, ablations, augmentation study	Done
ACCT development rounds 1–2 (EEGNet)	Done
Round 3: ACCT on ATCNet, TSception, EEG-TCNet, SpecNet	Running
Final held-out test (5 seeds, run once after design freeze)	Pending
Confirmation on a second dataset (e.g. COG-BCI)	Planned
Own recordings: Emotiv EPOC + Meta Quest 3S (after ethics approval)	Planned